$$\Delta E = 13.6 \times \left(\frac{1}{1} - \frac{1}{9}\right) \times 2^2$$

$$= 13.6\left(8/9\right) \times 2^2$$

(2)

$$K.E_1 = h\nu - h\nu_0$$

$$= 5 - 2 = 3$$

$$= 10 - 2 = 8$$

(20)

$$E = \frac{hc}{\lambda}$$

$E \qquad \frac{4E}{3} \qquad 2E$

$$\frac{4}{3}E - E$$

$$\frac{1}{3}E = \frac{hc}{\lambda_2}$$

$$\frac{1}{3}\frac{hc}{\lambda} = \frac{hc}{\lambda_2}$$

(21)

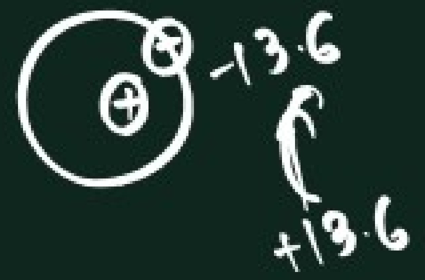
$$r = \frac{0.529 \times 16}{2} = 2.0$$

0.
9
5

$$\Delta E = +13.6 \times (1)\left(\frac{1}{1} - \frac{1}{\infty}\right) \qquad n_1 = 1$$

$$n_2 = \infty$$

$$= \underline{13.6\ eV}$$



Energy required to remove an electron = energy of that $e^{\ominus}$ in particular shell

$$\Delta E = h\nu \qquad \nu \approx \underline{10^{16}\ Hz} \text{ or } 9.8 \times 10^{15}\ Hz$$

$$\Delta E = \underline{2.18 \times 10^{-18}} \times Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\Delta E = 13.6 \times (1)\left(\frac{1}{1} - \frac{1}{n^2}\right)$$

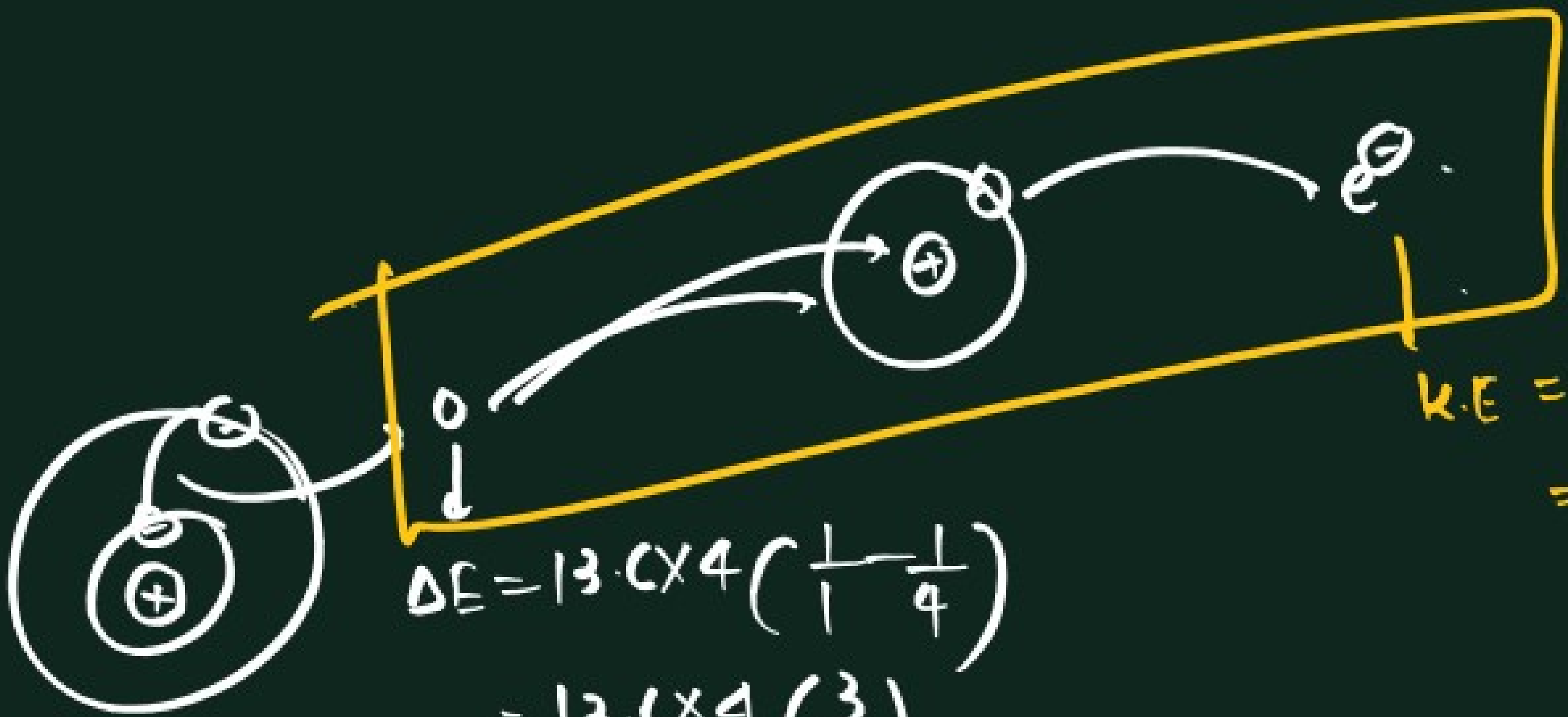
$$\frac{12.1}{13.6} = 1 - \frac{1}{n^2}$$

$$\frac{1}{n^2} = 1 - \frac{12.1}{13.6}$$

$$\frac{1}{n^2} = \frac{13.6 - 12.1}{13.6}$$

$$n^2 = \frac{13.6}{1.5} = 9 \qquad n^2 = 9$$

$$n = 3$$



$$\Delta E = 13.6 \times 4\left(\frac{1}{1} - \frac{1}{4}\right)$$

$$= 13.6 \times 4\left(\frac{3}{4}\right)$$

$$= 40.8\,eV.$$

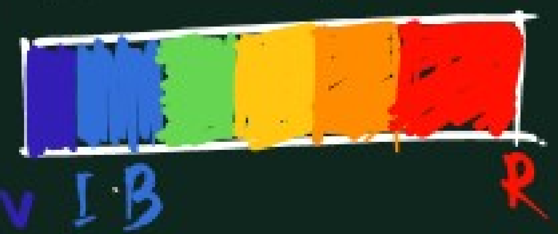
$$K.E = E - E_0$$

$$= 40.8 - 13.6 = \underline{27.2\,eV}$$

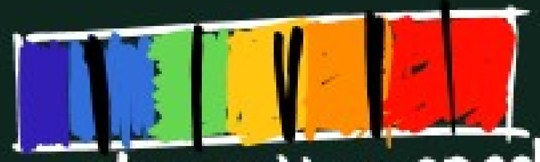
Hydrogen spectra →

spectra/ spectrum →

split pattern → spectrum



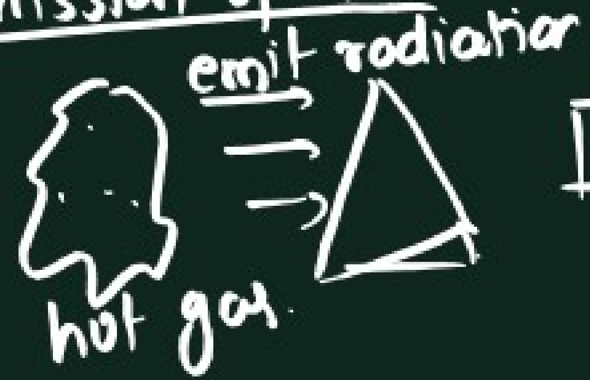
Absorption spectra →

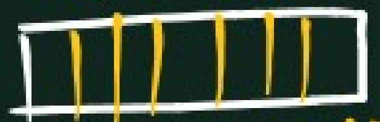


absorption spectra.

→ wavelength missing from spectra is absorb by gas

Emission spectra →

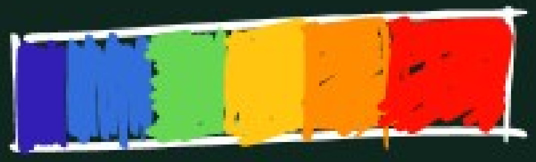




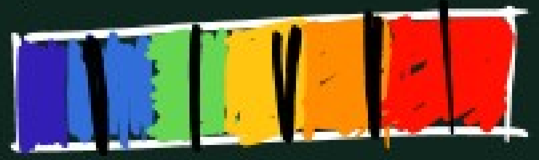
emission spectra ✓

spectrum

- atomic spectra
- molecular spectra
- line spectra
- band spectra
- continuous spectra



- discontinuous spectra.

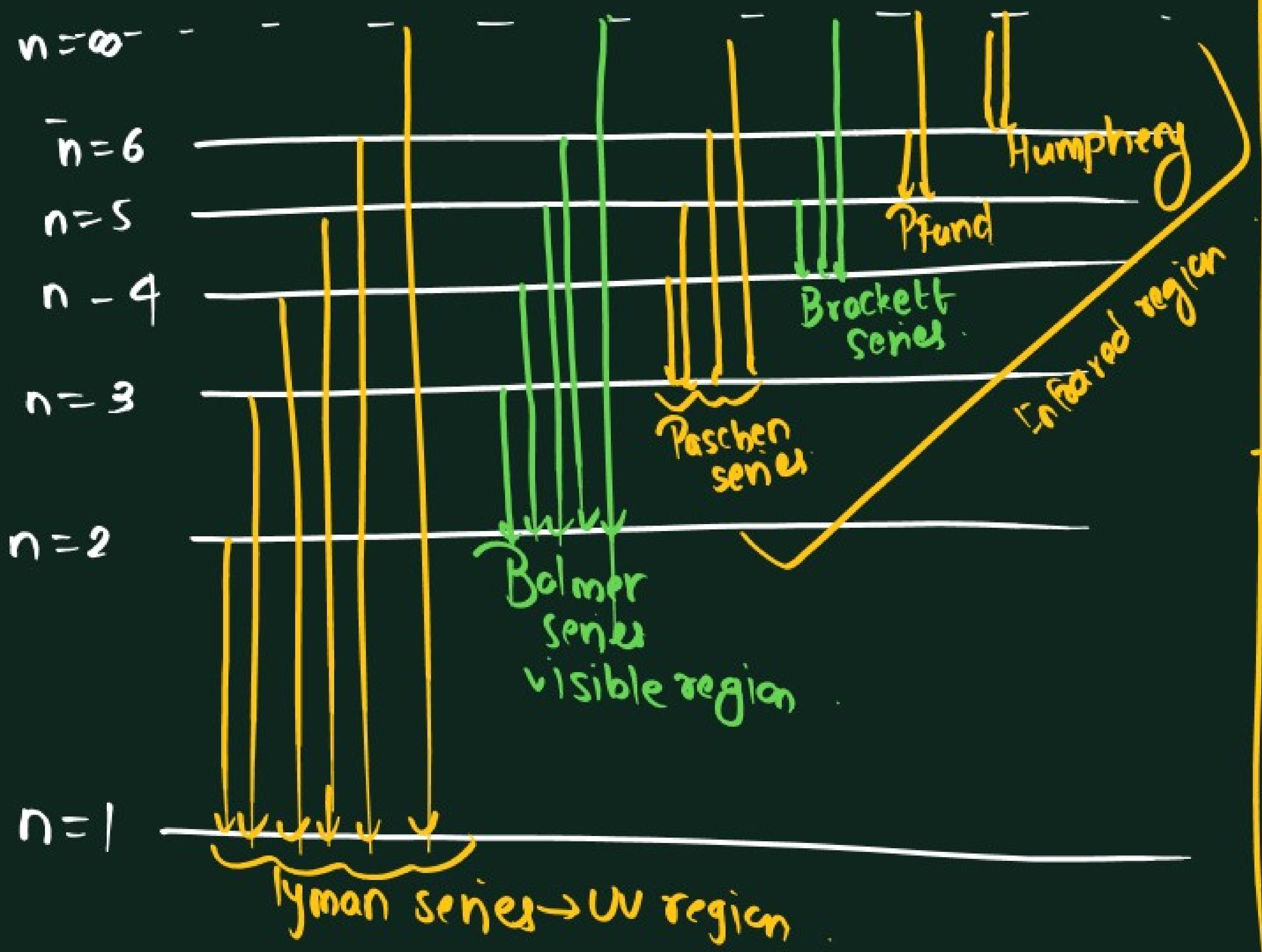


- absorption
- Emission spectra.

Hydrogen spectra →

→ emission spectra of hydrogen.

→ deexcitation of $e^{\ominus}$.



series	n_1	n_2
lyman	1	2,3,4 -- ∞
Balmer	2	3,4 --- ∞
Paschen	3	4,5 ... ∞
Brackett	4	5,6 -- ∞
Pfund	5	6,7 -- ∞
Humph..	6	7,8 -- ∞.

wavelength of any ~~line~~ spectral line →

$$\boxed{\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)}$$

R → Rydberg const → 109678 cm^{-1}

1st line

→ min Energy
→ min Freq^n
→ longest λ

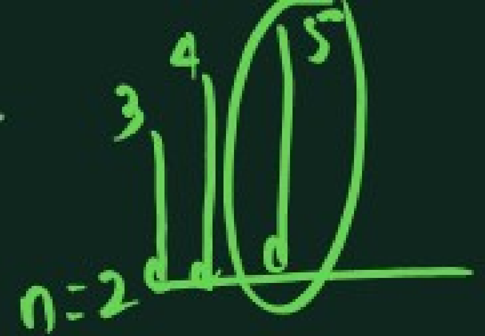
last line (ultimate or limiting line)

→ max^m energy

→ max^m ν
→ Shortest λ

For H atom
Balmer series → visible

1st line → Red
last line → violet



* 3rd line of Balmer series

$n_1 = 2$

$n_2 = x + 3 = 5$

(a) $n_1 = 2$ $z = 2$
$n_2 = 5$

$$\frac{1}{\lambda} = Rz^2\left(\frac{1}{4} - \frac{1}{25}\right)$$

$$\frac{1}{\lambda} = R \times \cancel{4}\left(\frac{21}{\cancel{4} \times 25}\right)$$

$$\boxed{\lambda = \frac{25}{21R}}$$

b) $n_1 = 3$
$n_2 = 3 + 1 = 4$
$\lambda = \frac{16}{7R}$

c) $\lambda = \frac{9}{8R}$

$n_1 = 1$
$n_2 = 3$

d) $\lambda = \frac{4}{R}$ $n_1 = 4$
$n_2 = \infty$

e) $\lambda = \frac{36}{5R}$ $n_1 = 2$ $n_2 = 3$

(f) last line
$n_1 = 5$
$n_2 = \infty$

$\lambda = \frac{25}{9R}$

(1) $\frac{\lambda_1}{\lambda_2} = \frac{16}{21}$

(2) Be^{+3}

$Z = 4$

$n_1 = 3$

$n_2 = 4$

λ

$\frac{1}{\lambda_1} = R16\left(\frac{1}{9} - \frac{1}{16}\right)$

$= R \times \cancel{16}\left(\frac{7}{\cancel{16} \times 9}\right)$

$\frac{1}{\lambda_1} = \frac{7R}{9}$

$\lambda_1 = \frac{9}{7R}$

He^{+}

$Z = 2$

λ_2

$n_1 = 2$

$n_2 = \infty$.

$\frac{1}{\lambda_2} = R\cancel{4} \times \left(\frac{1}{\cancel{4}}\right)$

$\lambda_2 = \frac{1}{R}$

$\frac{\lambda_1}{\lambda_2} = \frac{9/7\cancel{R}}{1/\cancel{R}} = 9/7$

(c) $\frac{\lambda_1}{\lambda_2} = \frac{36}{125}$.

(3) $n_1 = 3$ ✓

$n_2 = 6$ ✓

$Z = 3$

$\Delta E = 2.18 \times 10^{-18} \times Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$= 1.63 \times 10^{-18}\ J$

$\frac{1}{\lambda} = R\left(1 - \frac{1}{n^2}\right)$

$\frac{1}{\lambda R} = 1 - \frac{1}{n^2}$

$\frac{1}{n^2} = 1 - \frac{1}{\lambda R}$

$\frac{1}{n^2} = \frac{\lambda R - 1}{\lambda R}$

$n = \sqrt{\frac{\lambda R}{\lambda R - 1}}$

(4) A) $n_2 = \sqrt{\frac{\lambda R}{\lambda R + 1}}$

B) $n_2 = \sqrt{\frac{\lambda R - 1}{\lambda R}}$

C) $n_2 = \sqrt{\frac{\lambda R}{\lambda R - 1}}$

D) $n_2 = \sqrt{\frac{\lambda R}{1 - \lambda R}}$.

(5) ν_1

a) $\nu_3 = \nu_1 - \nu_2$

b) $\nu_3 = \nu_1 + \nu_2$

c) $\nu_3 = \nu_1 \times \nu_2$

d) $\nu_3 = \dfrac{1}{\nu_1 \times \nu_2}$

$$\frac{c}{\lambda_1} = CR\left(\frac{1}{1} - \frac{1}{\infty}\right)$$

$$\nu_1 = CR$$

$$\frac{1}{\lambda_2} = R\left(\frac{1}{1} - \frac{1}{4}\right)$$

$$\frac{c}{\lambda_2} = \frac{3RC}{4}$$

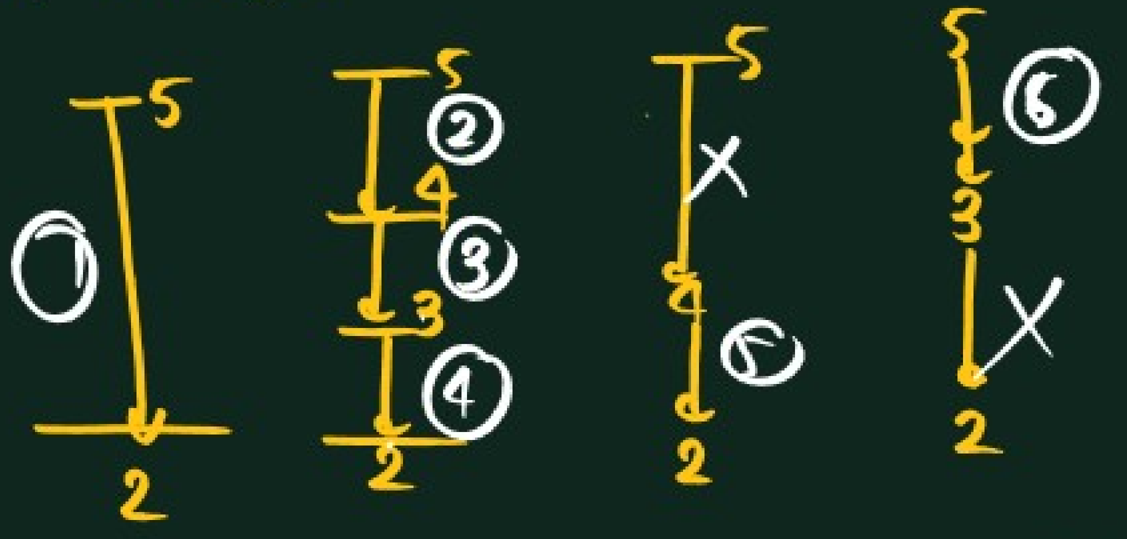
$$\nu_2 = \frac{3RC}{4}$$

$$\frac{1}{\lambda_3} = R\left(\frac{1}{4}\right)$$

$$\nu_3 = \frac{RC}{4}$$

* max^m no. of spectral lines possible (diff) $= \frac{(n_2-n_1)(n_2-n_1+1)}{2}$

* max^m No. of spectral lines possible during transition of $e^{\ominus}$ from 5 to 2 in H-atoms → 6



$= \Sigma\, n_2-n_1$

$= \Sigma\, 5-2$

$= \Sigma\, 3$

$= 3+2+1$

$= 6$

→ No. of visible lines → 3

→ max^m No. of spectral line possible during transition of $e^{\ominus}$ from 7 to 2 in a H-atom

$\boxed{\Sigma\, n_2-n_1}$

→ 5

Black body radiation →

light → ☐ → absorb
→ transmit
→ reflect

ordinary body

light → ☐ → absorb

black body

↳ absorbs light of all type of freq^n

→ Good absorber as well as good emitter.

→ classical approach about light

→ light has wave nature.

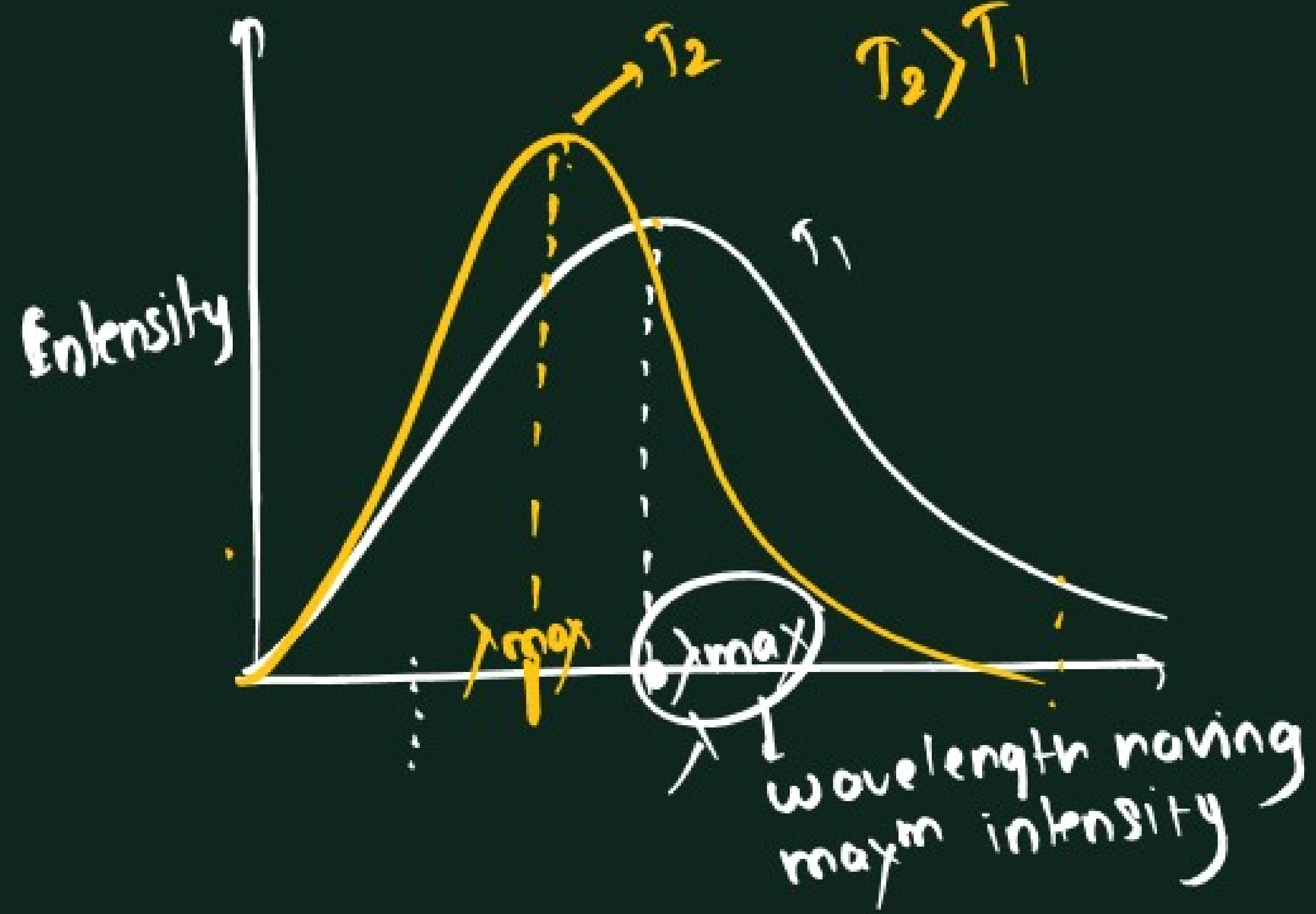
→ Energy $\propto$ Intensity ✓

& independent on ν & λ of light

* ☐ → emits radiation.

$T \uparrow$ Energy emitted $\uparrow$ $\nu \uparrow$

Red → orange → yellow → blue → violet



wein's displacement law →

$\lambda T = b \qquad b = 2.89 \times 10^{-3}\ mK.$

$$\boxed{\lambda \propto \frac{1}{T}}$$

* De-broglie Hypothesis → dual nature of matter.

→ light has dual nature → particle & wave.

→ matter → dual nature ?

→ The microscopic particle when travel with K.E they show wave characteristic.

→ matter which show waves are known as matter waves or de-broglie waves.

m → mass of particle (kg)
v → velocity –"–
K.E → K.E of –"–.

Davison & germer expt →

→ They obs. that beam of $e^{\ominus}$ shows diffraction phenomena.

→ diffraction shown by wave nature

$E = \frac{h\nu}{\lambda}$ $\quad$ $E = mv^2$

$\frac{h\nu}{\lambda} = mv^{\cancel{2}}$

$\boxed{\lambda = \frac{h}{mv}} = \frac{h}{p}$

de-broglie wavelength

As $m \uparrow$ $\lambda \downarrow$

→ for particle having high mass → λ is very small.

$K.E = \frac{1}{2}mv^2$

$m\,K.E = \frac{1}{2}(mv)^2$

$(mv)^2 = 2m\,K.E$

$mv = \sqrt{2mK.E}$

$\boxed{\lambda = \frac{h}{\sqrt{2mK.E}}}$

Intext → 4
Ex-1 → 23 – 36
Ex-2 → 22 - 30
Ex-3 → 3, 4, 7, 8, 10, 15
Ex-4 → 1, 3, 5, 7, 8
pyq → 10, 15, 16.